

Tower Maintenance Flow - From Start to Actual Task Build

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Source folder: /Users/ohadformanair/PycharmProjects/Tower_work/maintenance

Total maintenance tasks loaded	76
Non-draw-only tasks (planning cadence)	63
Purpose	Operational flow until building actual executable tasks

Flow Steps

- 1) Inventory baseline: fill stock, mounted quantity, location, serial, min level, and item type (Part/Tool/Consumable).
- 2) Clean task source: remove empty rows, validate Task_ID/Component/Task and cadence fields.
- 3) Builder setup: assign Task_Group(s), Tracking/Trigger mode(s), and timing intervals (hours/draw/calendar).
- 4) BOM build: define Required_Parts and Required_Tools from inventory list (not free-text only).
- 5) Work package: set Preparation, Safety protocol, Procedure, stop plan, estimated time, completion criteria.
- 6) Manual linkage: pin exact manual pages for task procedure and BOM parts context.
- 7) Pre-execution readiness: auto-check stock vs mounted and mark Ready / Missing / Wait-for-Part.
- 8) Order bridge: create part orders for missing consumables/parts before execution day.
- 9) Build actual task queue: from cadence/groups/schedule into executable day packs.
- 10) At this point tasks are ready for execution flow (start/done/history/report).

Quality Gate Before Execution

- Every task has Task_ID, Group(s), Trigger mode, and interval.
- Every task has BOM split into parts + tools.
- Safety protocol is present and high-fall-risk tasks force NO ENTRY.
- Missing parts generate order candidates before task execution.
- Manual context opens task procedure pages and relevant BOM pages.

Table: Tasks Not Only Per-Draw / Startup (Regular Cadence)

This table excludes draw-only startup tasks so planning can focus on weekly/monthly/3M/6M/hours/calendar/on-condition work.

#	Component	Task	Task ID	Groups	Mode	Hours Src	Interval	Required Parts	Source
1	63mm Furnace System	Cyclic purge after atmosphere exposure (minimum 5 cycles; end in vacuum)	FUR-MNT-012	Routine	event	nan	1 after exposure to atmosphere	Purge gas line flush consumables	Furnace63_Maintenance_Tracker_Template.xlsx
2	63mm Furnace System	Every 3 months OR 1500 operating hours: check pyrometer alignment & window cleanliness	FUR-MNT-007	3-Month	either	OperatingHours	3 months OR 1500 hours	Pyrometer window / cleaning kit; IPA wipes / cleaning cloth	Furnace63_Maintenance_Tracker_Template.xlsx
3	63mm Furnace System	Every 3 months OR 1500 operating hours: inspect graphite insulation/radiation shield; replace if needed	FUR-MNT-008	3-Month	either	OperatingHours	3 months OR 1500 hours	Graphite set / insulation parts	Furnace63_Maintenance_Tracker_Template.xlsx
4	63mm Furnace System	Every 6 months OR 3000 operating hours: check busbar overlap joints tightness	FUR-MNT-009	6-Month	either	OperatingHours	6 months OR 3000 hours		Furnace63_Maintenance_Tracker_Template.xlsx
5	63mm Furnace System	Every 6 months OR 3000 operating hours: inspect vacuum pump filter cleanliness	FUR-MNT-011	6-Month	either	OperatingHours	6 months OR 3000 hours	IPA wipes / cleaning cloth	Furnace63_Maintenance_Tracker_Template.xlsx
6	63mm Furnace System	Every 6 months OR 3000 operating hours: tower alignment check using plumb line	FUR-MNT-010	6-Month	either	OperatingHours	6 months OR 3000 hours		Furnace63_Maintenance_Tracker_Template.xlsx
7	63mm Furnace System	Monthly: inspect bottom doors for distortion/damage; replace if needed	FUR-MNT-006	Monthly	calendar	nan	1 months	Furnace bottom door set	Furnace63_Maintenance_Tracker_Template.xlsx
8	63mm Furnace System	On-condition: inspect/replace bottom sleeve top cap when eroded (e.g., length reduction >3 mm)	FUR-MNT-015	On-Condition	hours	OperatingHours	250 operating hours (typical)	Bottom sleeve top cap	Furnace63_Maintenance_Tracker_Template.xlsx
9	63mm Furnace System	On-condition: replace heating element when voltage trend increases (e.g., >2 V above new)	FUR-MNT-014	3-Month	hours	OperatingHours	500 operating hours (typical)	Heating element	Furnace63_Maintenance_Tracker_Template.xlsx
10	63mm Furnace System	Weekly: test water and gas flow interlocks	FUR-MNT-005	Weekly	calendar	nan	1 weeks	nan	Furnace63_Maintenance_Tracker_Template.xlsx
11	Bare Fibre Diameter Gauge	Align bare fibre diameter gauge to fibre line using tightened line	ALGN-003	Routine	event	nan	1 after gauge moved in	after re-threading / periodic verification	TowerAlignment_Maintenance_Tracker_Template.xlsx
12	Bottom Centering Pulley + Break Detector	Routine: ensure X/Y slides run free and true	BP-MNT-001	3-Month	calendar	nan	1 months	nan	BottomCenteringPulley_BreakDetector_Maintenance_Tracker_Template.xlsx
13	Bottom Centering Pulley + Break Detector	Weekly: check pulley rotates freely, bearing play, and clean V-groove	BP-MNT-002	Weekly, nan	calendar	nan	1 weeks	Bearing set; IPA wipes / cleaning cloth	BottomCenteringPulley_BreakDetector_Maintenance_Tracker_Template.xlsx

#	Component	Task	Task ID	Groups	Mode	Hours Src	Interval	Required Parts	Source
14	Cane / Rod Puller (Caterpillar)	Periodic: check carriage movement, belt tension, and lubricate drive linkages	CP-MNT-002	Routine	calendar	nan	1 months	Light machine oil	CanePuller_Maintenance_Tracker_Template.xlsx
15	Cane Puller	Align cane puller belts to fibre line; record micrometer baseline and offset for cane diameter	ALGN-006	Routine	event	nan	1 after puller moved / 1 after belt change / when drawing cane	LowerAlignment_Maintenance_Tracker_Template.xlsx	
16	Capstan	Set capstan offset (nominal 50 mm) and mark/bolt capstan position	ALGN-002	Routine	event	nan	1 after capstan moved / after establishing fibre line	TowerAlignment_Maintenance_Tracker_Template.xlsx	
17	Capstan System	As needed: adjust urethane belt tension (cam plate)	CAP-MNT-003	On-Condition	event	nan	1 when slipping / after belt change	Drive belt	Capstan_Maintenance_Tracker_Template.xlsx
18	Capstan System	Every 3 months: check bearings are not worn (sealed for life)	CAP-MNT-002	3-Month	calendar	nan	3 months	Bearing set	Capstan_Maintenance_Tracker_Template.xlsx
19	Clean Air System (HEPA + Pre-filter + Fan Cabinet)	As needed: replace HEPA filter element (front face access; tighten nuts diagonally)	CAS-MNT-006	On-Condition	event	nan	1 when airflow incorrect / after filter change	Pre-filter element; Pre-filter seal/gasket; HEPA filter element; HEPA gasket/seal; IPA wipes / cleaning cloth	CleanAirSystem_Maintenance_Tracker_Template.xlsx
20	Clean Air System (HEPA + Pre-filter + Fan Cabinet)	As needed: replace pre-filter element (with seal inspection and correct airflow direction)	CAS-MNT-005	On-Condition	event	nan	1 when pre-filter clogged / after filter change	Pre-filter element; Pre-filter seal/gasket; HEPA filter element; HEPA gasket/seal; IPA wipes / cleaning cloth	CleanAirSystem_Maintenance_Tracker_Template.xlsx
21	Clean Air System (HEPA + Pre-filter + Fan Cabinet)	Every 6 months: replace pre-filter; re-adjust speed control; replace HEPA if airflow remains incorrect	CAS-MNT-004	6-Month	calendar	nan	6 months	Pre-filter element; Pre-filter seal/gasket; HEPA filter element; HEPA gasket/seal; IPA wipes / cleaning cloth	CleanAirSystem_Maintenance_Tracker_Template.xlsx
22	Clean Air System (HEPA + Pre-filter + Fan Cabinet)	Monthly: inspect wear/damage, verify intakes clear, verify airflow uniform & at optimum; balance dampers	CAS-MNT-003	Monthly	calendar	nan	1 months	Pre-filter element; Pre-filter seal/gasket; IPA wipes / cleaning cloth	CleanAirSystem_Maintenance_Tracker_Template.xlsx
23	Clean Air System (HEPA + Pre-filter + Fan Cabinet)	When switching on: confirm fibre position in bare fibre gauge window stays centered; correct if needed	CAS-MNT-002	Routine	event	nan	1 when fan comes on / after filter change	Pre-filter element; Pre-filter seal/gasket; IPA wipes / cleaning cloth	CleanAirSystem_Maintenance_Tracker_Template.xlsx
24	Coated Fibre Diameter Gauge	Align coated fibre diameter gauge to fibre line using tightened line	ALGN-010	Routine	event	nan	1 after gauge moved / in periodic verification		TowerAlignment_Maintenance_Tracker_Template.xlsx
25	Coating System (Resin Delivery Block)	Align coating resin delivery block to fibre line; record X/Y micrometer readings	ALGN-007	Routine	event	nan	1 after coating head moved / after resin change	Resin line/solvent; Application solvent	TowerAlignment_Maintenance_Tracker_Template.xlsx

#	Component	Task	Task ID	Groups	Mode	Hours Src	Interval	Required Parts	Source
26	Concentricity Monitor	Align concentricity monitor optics and laser aim to fibre line; fine-tune during draw	ALGN-008	Routine	event	nan	1 after monitor moved	nan / after optics service	TowerAlignment_Maintenance_Tracker_Template.xlsx
27	Draw Tower – Alignment (overall)	Initial setup: establish fibre line using plumb line through furnace to datum pulley	ALGN-001	Routine	event	nan	1 first tower erection	nan after furnace moved	TowerAlignment_Maintenance_Tracker_Template.xlsx
28	Draw Tower – Alignment (overall)	Periodic verification: check fibre line alignment with tightened line through key stations	ALGN-011	6-Month	calendar	nan	6 months	nan	TowerAlignment_Maintenance_Tracker_Template.xlsx
29	Dual Drum Winder	3■monthly: grease drum bearing housings	LUB-DDW-002	3-Month	calendar	nan	3 months	Grease (NLGI-2)	GenericLubrication_Maintenance_Tracker_Template.xlsx
30	Dual Drum Winder	Monthly: lubricate clamps, guide pulleys, layering arm pulleys	LUB-DDW-001	Monthly	calendar	nan	1 months	Light machine oil	GenericLubrication_Maintenance_Tracker_Template.xlsx
31	General Tower Components	Daily: cleanliness, smooth operation, pulley condition	LUB-GEN-001	Daily	calendar	nan	1 days	IPA wipes / cleaning cloth	GenericLubrication_Maintenance_Tracker_Template.xlsx
32	General Tower Shafts & Slideways	Monthly: wipe bright shafts and slideways with light oil	LUB-GEN-002	Monthly	calendar	nan	1 months	Light machine oil	GenericLubrication_Maintenance_Tracker_Template.xlsx
33	Guide Pulley + Fibre Tension Gauge	Monthly (or as required): check tension gauge calibration	TG-MNT-002	Monthly, nan	calendar	nan	1 months	nan	GuidePulley_TensionGauge_Maintenance_Tracker_Template.xlsx
34	Guide Pulley + Fibre Tension Gauge	On-condition: return load cell to manufacturer if erratic / cannot calibrate	TG-MNT-003	On-Condition, nan	event	nan	1 if will not calibrate	load cell service/replacement	GuidePulley_TensionGauge_Maintenance_Tracker_Template.xlsx
35	Guide Pulley + Fibre Tension Gauge	Weekly: check pulley rotation, bearing play, and clean V-groove	TG-MNT-001	Weekly, nan	calendar	nan	1 weeks	Bearing set, IPA wipes - small	GuidePulley_TensionGauge_Maintenance_Tracker_Template.xlsx
36	He Cooling System	Align He cooling tube entry/exit orifices to fibre line; balance ram stops	ALGN-004	Routine	event	nan	1 after cooling tube removed	nan / after maintenance	TowerAlignment_Maintenance_Tracker_Template.xlsx
37	Helium Cooling System	Monthly: lubricate slide bearings on cooling tubes	LUB-HCS-001	Monthly	calendar	nan	1 months	Light machine oil	GenericLubrication_Maintenance_Tracker_Template.xlsx

#	Component	Task	Task ID	Groups	Mode	Hours Src	Interval	Required Parts	Source
38	Linear Dancer	Monthly: lubricate dancer pulley spindles and bearings	LUB-LD-001	Monthly	calendar	nan	1 months	Light machine oil	GenericLubrication_Maintenance_Tracker_Template.xlsx
39	Mini Capstan	Align mini capstan pinch roller to fibre line (coarse + fine adjustment)	ALGN-005	Routine	event	nan	1 after mini capstan removed / after roller service		TowerAlignment_Maintenance_Tracker_Template.xlsx
40	Mini Capstan	Weekly: inspect cleanliness, roller bearings, carriage bearings, glass embedment, lubricate, check pneumatic leaks	MC-MNT-002	Weekly	calendar	nan	1 weeks	Light machine oil; IPA wipes / cleaning cloth	MiniCapstan_Maintenance_Tracker_Template.xlsx
41	Mini Capstan	Weekly: lubricate carriage guides and roller bearings	LUB-MCAP-001	Weekly	calendar	nan	1 weeks	Light machine oil	GenericLubrication_Maintenance_Tracker_Template.xlsx
42	Preform Feed Assembly	3-monthly: grease ball screw	PF-MNT-005	3-Month	calendar	nan	3 months	Grease (NLGI-2)	PreformFeedAssembly_Maintenance_Tracker_Template.xlsx
43	Preform Feed Assembly	3-monthly: grease linear rails	PF-MNT-004	3-Month	calendar	nan	3 months	Grease (NLGI-2)	PreformFeedAssembly_Maintenance_Tracker_Template.xlsx
44	Preform Feed Assembly	3-monthly: lubricate X-Y slide bearings	PF-MNT-003	3-Month	calendar	nan	3 months	Light machine oil	PreformFeedAssembly_Maintenance_Tracker_Template.xlsx
45	Preform Feed Assembly	3-monthly: lubricate ball screw support bearings	PF-MNT-006	3-Month	calendar	nan	3 months	Light machine oil	PreformFeedAssembly_Maintenance_Tracker_Template.xlsx
46	Preform Feed Assembly	3-monthly: lubricate clamp moving parts	PF-MNT-002	3-Month	calendar	nan	3 months	Light machine oil	PreformFeedAssembly_Maintenance_Tracker_Template.xlsx
47	Preform Feed Assembly	Daily: inspect cleanliness and smooth operation of clamp and X-Y slides	PF-MNT-001	Daily	calendar	nan	1 days	IPA wipes / cleaning cloth	PreformFeedAssembly_Maintenance_Tracker_Template.xlsx
48	Preform Feed Module	3-monthly: grease linear bearings and ball screw (if required)	LUB-PFM-001	3-Month	calendar	nan	3 months	Grease (NLGI-2)	GenericLubrication_Maintenance_Tracker_Template.xlsx
49	Preform Feed Module	6-monthly: lubricate actuator O-ring seals	LUB-PFM-002	6-Month	calendar	nan	6 months	Light machine oil	GenericLubrication_Maintenance_Tracker_Template.xlsx

#	Component	Task	Task ID	Groups	Mode	Hours Src	Interval	Required Parts	Source
50	Pressurised Coating System	Monthly: lubricate applicator gimbals and X■Y adjustment mechanism	LUB-PCS-001	Monthly	calendar	nan	1 months	Applicator seal kit; Light machine oil	GenericLubrication_Maintenance_Tracker_Template.xlsx
51	Take Up Winder	Every 3 months: grease rotating shaft; wipe & grease guide rails (thin film only)	TUW-MNT-002	3-Month	calendar	nan	3 months	Grease (NLGI-2)	TakeUpWinder_Maintenance_Tracker_Template.xlsx
52	Take Up Winder	Every 6 months: inspect drive train (belt tension + contamination/wear/damage)	TUW-MNT-003	6-Month	calendar	nan	6 months	Drive belt	TakeUpWinder_Maintenance_Tracker_Template.xlsx
53	UV Curing System	After full preform draw: inspect quartz liner tube for discolouration (replace if needed)	UV-MNT-002	Routine	event	nan	1 after full preform draw	Quartz liner tube	UV_CuringSystem_Maintenance_Tracker_Template.xlsx
54	UV Curing System	Align UV lamp entry/exit irises to fibre line (hexapod turnbuckles)	ALGN-009	Routine	event	nan	1 after UV lamp moved / after maintenance		TowerAlignment_Maintenance_Tracker_Template.xlsx
55	UV Curing System	Every 500 operating hours: clean lamp module, reflector, bulb and RF screen	UV-MNT-003	3-Month	hours	lamp_hours	500 hours	UV lamp cleaning kit; Optics wipes; IPA wipes / cleaning cloth	UV_CuringSystem_Maintenance_Tracker_Template.xlsx
56	UV Curing System	Monthly: lubricate UV lamp VAM pivot hinges	LUB-UV-001	Monthly	calendar	nan	1 months	Light machine oil	GenericLubrication_Maintenance_Tracker_Template.xlsx
57	UV Curing System – PSU	Every 3 months: remove dust from PSU units and replace air filters if required	UV-MNT-004	3-Month	calendar	nan	3 months	Air filter element	UV_CuringSystem_Maintenance_Tracker_Template.xlsx
58	Wet-on-Dry Coating System	Applicator alignment to tower axis check	WOD-MNT-008	Routine	calendar	nan	1 months	Applicator seal kit	WOD_Maintenance_Tracker_Template.xlsx
59	Wet-on-Dry Coating System	Applicator/delivery block levelness check	WOD-MNT-005	Routine	calendar	nan	1 days	Applicator seal kit	WOD_Maintenance_Tracker_Template.xlsx
60	Wet-on-Dry Coating System	End-of-draw housekeeping	WOD-MNT-004	Routine	event	nan	1 end of drawing	nan	WOD_Maintenance_Tracker_Template.xlsx
61	Wet-on-Dry Coating System	Resin line purge after refill / dip-tube exposure	WOD-MNT-002	Routine	event	nan	1 after refill	Resin line seals; Resin line cleaning solvent; Purge gas / line flush consumables	WOD_Maintenance_Tracker_Template.xlsx
62	Wet-on-Dry Coating System	Water circuit checks (weekly) + flush (annual)	WOD-MNT-007	Weekly	calendar	nan	1 weeks	nan	WOD_Maintenance_Tracker_Template.xlsx

#	Component	Task	Task ID	Groups	Mode	Hours Src	Interval	Required Parts	Source
63	Wet-on-Dry Coating System	Weekly: joints leakage check + line purge + cleaning	WOD-MNT-006	Weekly	calendar	nan	1 weeks	IPA wipes / cleaning cloth; Purge gas / line flush consumables	WOD_Maintenance_Tracker_Template.xlsx